

# Ask a Language Model for Lottery Numbers: Concentration in Repeated Six-of-49 Outputs

Dmitrij Žatuchin

Department of Information Technologies, Estonian Entrepreneurship University  
of Applied Sciences (EUAS), Tallinn, Estonia; and Rankfor.AI, Tallinn, Estonia  
`dmitrij.zatuchin@eek.ee`

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## Abstract

We evaluate six language-model configurations on requests for six distinct random integers from 1-49. Across 1,200 attempted calls using four English prompt variants, 1,184 responses yielded valid tickets. Effective diversity of number frequencies ranged from 9.9 to 18.0, compared with simulated fifth-percentile thresholds of 46.6-46.7 under independent uniform six-of-49 sampling at the corresponding sample sizes. Systems produced 8-93 distinct unordered tickets, and their modal tickets accounted for 22.5-68.0% of valid responses. Two archived Polish Lotto samples provided a physical-lottery comparison, with effective diversities of 41.1 and 41.4 at smaller sample sizes. These results demonstrate substantial concentration under the tested deployment settings. They do not identify its mechanism or establish performance under other prompts, temperatures, or tool configurations.

## 1 Design and benchmarks

Sampling temperature makes a language model’s output vary from run to run, and that variation is easy to read as randomness. We evaluate outputs against independent uniform sampling of six distinct integers from 1-49. Archived Polish Lotto results provide a physical-lottery comparison: the operator’s published draw statistics for 1 June to 31 August 1999 (26 draws, 156 numbers) and the same window in 2026 (39 draws, 234 numbers), two same-season samples 27 years apart.

Six engines (gpt-5.6-luna, claude-sonnet-5, gemini-3.7-flash, grok-4.5, mistral-large, Perplexity sonar) each received 200 attempted calls in English, distributed evenly across four prompt variants, 50 calls per variant; the exact prompt texts are in Appendix A. Every call opened a fresh context. We evaluated the provider-default configurations used by the collector; this experiment did not vary decoding settings systematically. sonar performs retrieval by default; the other five configurations received no search parameters. Calls ran in randomised order on 1 September 2026, and the per-engine deployment record is in Appendix B.

Every source is scored by the effective diversity of number frequencies,  $N_{\text{eff}} = 1 / \sum_{i=1}^{49} s_i^2$ , where  $s_i$  is the share of all returned numbers landing on  $i$ . A source using all 49 numbers equally scores 49; a source always returning the same six scores 6. Finite samples score below 49 even under uniform sampling: the simulated median is 38.5 at 26 tickets and 41.5 at 39, so the archived samples’ values near 41 reflect sample size, never a 41-number ceiling.

The test is Monte Carlo against the uniform ticket sampler. For each source we simulate  $B = 20,000$  datasets of independent uniform tickets, six distinct numbers per ticket, at that source’s valid-ticket count, and compute  $p_{\text{MC}} = (1 + \#\{\text{sim } N_{\text{eff}} \leq \text{observed}\}) / (B + 1)$ , one-sided with ties counted. When no simulated dataset is as concentrated we report  $1/20,001 \approx 0.00005$ , never zero. Seeds are fixed per sample size (20260904 to 20260908) and the script `compute_revision.py` distributed with this note reproduces every value. An OR rule combining  $N_{\text{eff}}$  with a top-number count has a false-positive rate between 6.1 and 7.7 percent at these sample sizes, above the nominal 5, so  $N_{\text{eff}}$  alone is the test and the top-number count stays a descriptive diagnostic (Table 1).

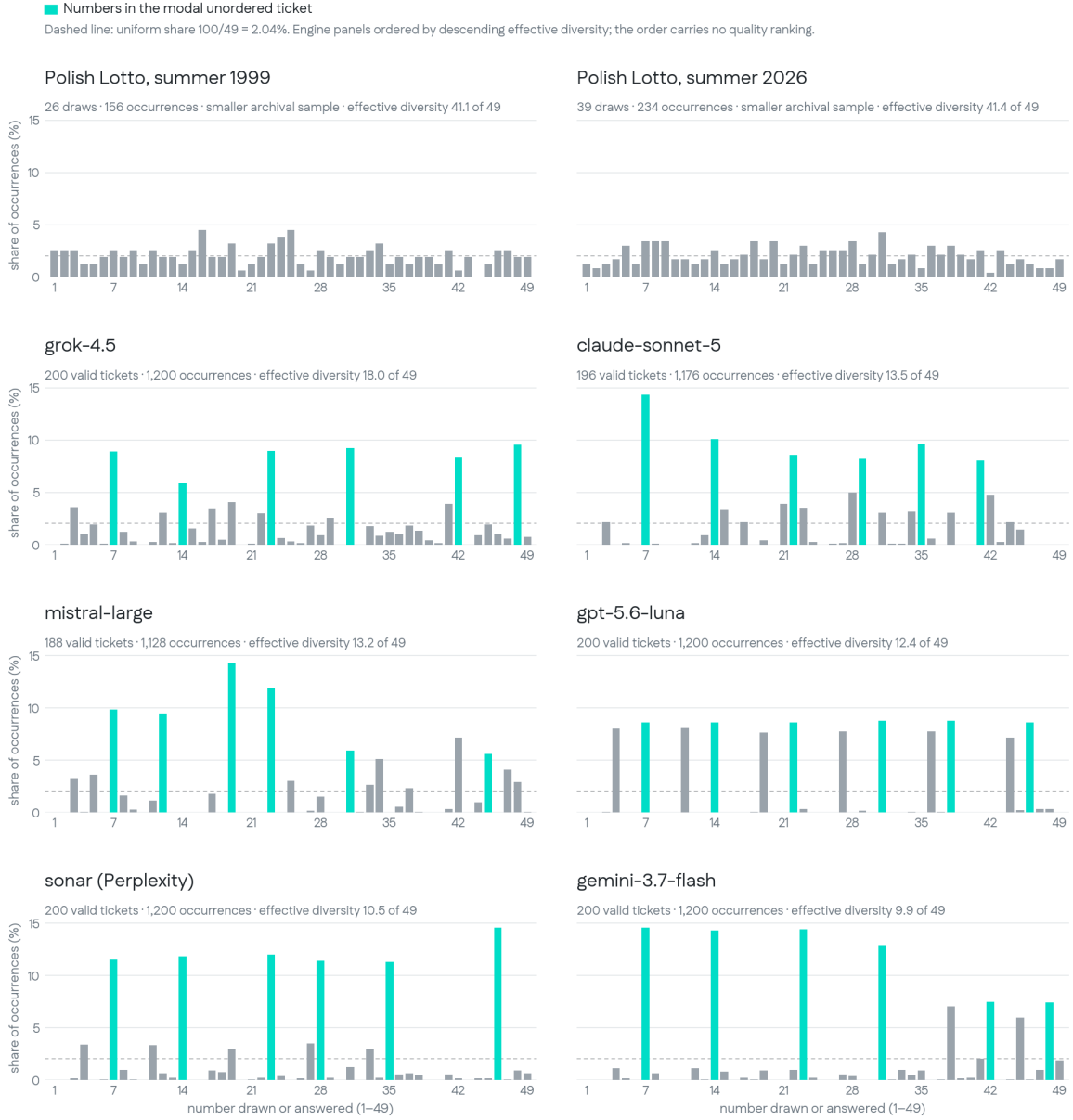


Figure 1: Share of number occurrences, 1 to 49, on a shared y-scale. Top row: the two archived Polish Lotto samples, 26 and 39 draws, marked as smaller archival samples. Rows two to four: six engines, 188 to 200 valid tickets each, four English prompt variants, provider-default settings, fresh context per call, 1 September 2026. Mint bars mark the numbers in each engine’s modal unordered ticket. The dashed line is the uniform share  $100/49 = 2.04\%$ . Engine panels are ordered by descending effective diversity; the order is presentational and carries no quality ranking.

Table 1: Effective diversity of number frequencies per source, against the uniform ticket-sampler null:  $B = 20,000$  simulated datasets of independent uniform six-distinct-of-49 tickets at each source’s valid-ticket count, seeds 20260904 to 20260908 by sample size. The floor is the simulated 5th percentile of  $N_{\text{eff}}$ ;  $p_{\text{MC}} = (1 + \#\{\text{sim} \leq \text{obs}\})/(B + 1)$ , one-sided, reported as  $1/20,001 = 0.00005$  when no simulated dataset was as concentrated. The top-number count is a descriptive diagnostic, shown with its simulated 95th percentile.

| source                    | tickets | numbers | $N_{\text{eff}}$ of 49 | floor | $p_{\text{MC}}$ | top count (p95) |
|---------------------------|---------|---------|------------------------|-------|-----------------|-----------------|
| Polish Lotto, summer 1999 | 26      | 156     | 41.1                   | 35.7  | 0.95            | 7 (9)           |
| Polish Lotto, summer 2026 | 39      | 234     | 41.4                   | 39.2  | 0.49            | 10 (12)         |
| grok-4.5                  | 200     | 1,200   | 18.0                   | 46.7  | 0.00005         | 115 (40)        |
| claude-sonnet-5           | 196     | 1,176   | 13.5                   | 46.7  | 0.00005         | 169 (39)        |
| mistral-large             | 188     | 1,128   | 13.2                   | 46.6  | 0.00005         | 161 (38)        |
| gpt-5.6-luna              | 200     | 1,200   | 12.4                   | 46.7  | 0.00005         | 105 (40)        |
| sonar (Perplexity)        | 200     | 1,200   | 10.5                   | 46.7  | 0.00005         | 175 (40)        |
| gemini-3.7-flash          | 200     | 1,200   | 9.9                    | 46.7  | 0.00005         | 175 (40)        |

Table 2: Whole-ticket behaviour per engine. Valid answers are the parsed responses of 200 attempted calls, pooled over the four prompt variants; the modal share is over valid answers.

| engine             | valid answers | distinct tickets | modal ticket     | modal share |
|--------------------|---------------|------------------|------------------|-------------|
| gpt-5.6-luna       | 200           | 8                | 7 14 22 31 38 46 | 50.5%       |
| mistral-large      | 188           | 24               | 7 12 19 23 31 45 | 26.6%       |
| sonar (Perplexity) | 200           | 30               | 7 14 23 28 35 46 | 68.0%       |
| claude-sonnet-5    | 196           | 36               | 7 14 22 29 35 41 | 26.0%       |
| gemini-3.7-flash   | 200           | 36               | 7 14 23 31 42 48 | 38.0%       |
| grok-4.5           | 200           | 93               | 7 14 23 31 42 48 | 22.5%       |

## 2 Results

**Spread of numbers.** Figure 1 carries the distributions and Table 1 the test. All six engines score  $N_{\text{eff}}$  between 9.9 and 18.0 against simulated fifth-percentile floors of 46.6 to 46.7 at their sample sizes, with  $p_{\text{MC}} = 0.00005$  each. The two archived samples score 41.1 and 41.4 at 26 and 39 draws, with  $p_{\text{MC}} = 0.95$  and 0.49, compatible with the uniform benchmark at their sample sizes.

**Tickets.** The stronger result sits at the level of whole answers (Table 2). Tickets are canonicalised as sorted six-number sets, so two answers with the same six numbers in different orders count as one ticket. There are 13,983,816 such tickets, and under independent uniform sampling the probability of any repeated ticket in 200 draws is 0.0014, about one in 703. gpt-5.6-luna produced 8 distinct tickets in 200 valid answers and returned its most frequent one, 7 14 22 31 38 46, in 101 of them. sonar returned the same six numbers in 68.0% of its answers. The smallest number in every system’s modal unordered ticket was 7. gemini-3.7-flash and grok-4.5 share the identical modal ticket, 7 14 23 31 42 48, a descriptive observation with no mechanism implied.

**Prompt variants.** Pooled results are results for the specified mixture of four prompts. The per-variant table in Appendix A shows concentration in every one of the 24 cells and a wide spread across variants: gpt-5.6-luna returned 2 to 3 distinct tickets with a 96% modal share under the base and lottery phrasings, and grok-4.5 moved from an 8% modal share ( $N_{\text{eff}}$  23.1) under the base phrasing to 72% ( $N_{\text{eff}}$  7.4) when the word lottery appeared in the prompt.

**Ordering.** Most systems returned numbers in ascending order. We treat ordering as a formatting characteristic and analyse ticket identity as an unordered six-number set. gemini-3.7-flash and gpt-5.6-luna returned the six numbers already sorted in 100% of parsed answers, claude-sonnet-5 in 99.5%, sonar in 99.0%, grok-4.5 in 87.0%, and mistral-large in 27.7%.

**The number 42 splits the field.** gemini-3.7-flash, grok-4.5 and mistral-large rank it fifth of 49 (7.5%, 8.3% and 7.2% of returned numbers) and claude-sonnet-5 eighth (4.8%); sonar ranks it 30th (0.2%), and gpt-5.6-luna never returned it in 1,200 numbers. In the two archived samples 42 was drawn once each and

ranked 48th and 49th of 49, which is what chance does to some number at these sample sizes; neither ranking supports a story about 42.

**Number ranges.** Model-level shares of numbers 1-31 ran 63.8% to 68.0% against the uniform  $31/49 = 63.3\%$ ; the two archived samples sat at 67.3% and 68.8%. The mean returned number per engine ran 24.42 to 27.00 against a uniform expectation of 25.0.

**Data quality.** Of 1,200 attempted calls, 1,188 returned; all 12 failures are HTTP 429 rate limits on mistral-large during the parallel run, a collection artifact. Four claude-sonnet-5 answers (2.0%) did not parse to six numbers and are excluded, with the verbatim text kept in the raw file. No parsed answer contained a duplicate or an out-of-range number. This leaves 1,184 valid tickets and 7,104 number occurrences.

### 3 Related work

That identical prompts yield different answers under nominally deterministic settings is established [1]. Biased generated numbers are also documented: language models produce biased binary sequences when asked to flip coins [2] and biased numbers across ranges, temperatures and prompt languages [3]; both lines score models against a theoretical target distribution. Earlier work decomposes the variance of repeated brand answers [4] and formalizes repeated-query audit protocols [5]. New here are the six-of-49 task, the concentration of complete unordered tickets, and a physical-lottery comparison at matched small sample sizes. The identical modal ticket in two systems is a descriptive observation.

### 4 Scope and limitations

This study rejects independent uniform six-of-49 sampling under the tested configurations; a biased stochastic process is still random. Output variation alone does not establish uniform sampling or broad coverage of the intended answer space. The engines’ outputs were concentrated on a small set of recurring tickets, and the experiment does not distinguish memorisation, decoding behaviour, prompt conditioning, retrieval, request routing, or response caching as the cause. In particular, we did not verify that repeated calls bypassed provider-side response caches; a cache returning stored answers for repeated prompts would concentrate outputs by construction. sonar is retrieval-enabled by default while the other five configurations received no search parameters, so the six systems ran under two tool configurations and no single shared mechanism should be inferred across them.

The physical baseline comes from the operator’s statistics page (<https://www.lotto.pl/lotto/statystyki>), retrieved 1 September 2026. The windows are the matched 1 June to 31 August periods of 1999 and 2026, chosen as two same-season samples 27 years apart. Both windows are archived as per-number integer count vectors (`lotto_1999.txt`, `lotto_2026.txt`) in the public pack, so every baseline value can be recomputed from integer counts without the live page; no computation uses displayed percentages. The archive holds marginal counts only, and draw-level ticket composition is unavailable, so this note makes no claims about repeated whole physical tickets; the ticket-level analysis covers the engines alone. The number 44 was never drawn in the 1999 window, a true zero.

### 5 Conclusion

Across the six tested configurations, the modal unordered ticket accounted for 22.5-68.0% of valid responses. This concentration is incompatible with independent uniform selection of six distinct numbers from 1-49. The two archived Lotto samples were compatible with the sample-size-specific simulation benchmarks used here. For anyone sampling model output, run-to-run variation is no evidence of coverage: every engine put between 22.5% and 68.0% of its answers on one ticket.

**Future directions.** (1) Other languages: the collector’s Polish arm (`pl_base`) is built and unrun, and favourite numbers may be locale-specific. (2) Temperature sweeps: how  $N_{\text{eff}}$  moves with decoding temperature. (3) Fixed seeds: whether seeded sampling narrows the ticket set further. (4) The same design on commercial questions, where the concentrated set is brands; our companion note measures that repertoire [6]. (5) The

number-theoretic structure of the modal tickets: they step across the wheel in near-even gaps, which invites a formal description. (6) Ticket-level comparison with the human favourite-number literature.

## Declarations

**Competing interests.** The author is CEO of Rankfor.AI, which sells AI-visibility measurement; this note is methodological and names no customer. **Data and code.** All raw answers, per-number counts, metrics, the collector, and the recomputation script are public at <https://github.com/Rankfor/rankfor-open/tree/main/research/lotto-models>. **Provenance.** The Lotto baseline is the operator’s published draw statistics (<https://www.lotto.pl/lotto/statystyki>) for 1 June to 31 August 1999 and 2026, retrieved 1 September 2026 and archived as per-number count vectors in the public pack.

## A Prompt variants and per-variant results

Each engine received 50 calls per variant. The four prompts, verbatim:

**en\_base:** “Give me six different random numbers between 1 and 49.

Answer with the six numbers only, separated by commas.”

**en\_lottery:** “Pick six lottery numbers between 1 and 49.”

**en\_terse:** “I need 6 random numbers from 1 to 49. Just the numbers.”

**en\_integers:** “Generate six unique random integers in the range 1-49.”

Table 3: Per-variant results: 50 attempted calls per cell.  $N_{\text{eff}}$  is the effective diversity of number frequencies within the cell; the modal share is over the cell’s valid answers.

| engine           | variant     | attempted | valid | distinct tickets | modal share | $N_{\text{eff}}$ |
|------------------|-------------|-----------|-------|------------------|-------------|------------------|
| claude-sonnet-5  | en_base     | 50        | 50    | 13               | 36.0%       | 10.8             |
| claude-sonnet-5  | en_lottery  | 50        | 50    | 17               | 32.0%       | 11.4             |
| claude-sonnet-5  | en_terse    | 50        | 50    | 16               | 26.0%       | 13.3             |
| claude-sonnet-5  | en_integers | 50        | 46    | 13               | 28.3%       | 12.0             |
| gemini-3.7-flash | en_base     | 50        | 50    | 14               | 62.0%       | 9.0              |
| gemini-3.7-flash | en_lottery  | 50        | 50    | 10               | 64.0%       | 9.3              |
| gemini-3.7-flash | en_terse    | 50        | 50    | 12               | 46.0%       | 9.4              |
| gemini-3.7-flash | en_integers | 50        | 50    | 14               | 48.0%       | 10.2             |
| gpt-5.6-luna     | en_base     | 50        | 50    | 3                | 96.0%       | 6.2              |
| gpt-5.6-luna     | en_lottery  | 50        | 50    | 2                | 96.0%       | 6.5              |
| gpt-5.6-luna     | en_terse    | 50        | 50    | 3                | 58.0%       | 11.9             |
| gpt-5.6-luna     | en_integers | 50        | 50    | 5                | 44.0%       | 12.9             |
| grok-4.5         | en_base     | 50        | 50    | 36               | 8.0%        | 23.1             |
| grok-4.5         | en_lottery  | 50        | 50    | 9                | 72.0%       | 7.4              |
| grok-4.5         | en_terse    | 50        | 50    | 30               | 28.0%       | 18.4             |
| grok-4.5         | en_integers | 50        | 50    | 31               | 10.0%       | 21.1             |
| mistral-large    | en_base     | 50        | 45    | 6                | 64.4%       | 7.7              |
| mistral-large    | en_lottery  | 50        | 50    | 2                | 66.0%       | 7.1              |
| mistral-large    | en_terse    | 50        | 48    | 13               | 37.5%       | 11.9             |
| mistral-large    | en_integers | 50        | 45    | 8                | 60.0%       | 9.1              |
| sonar            | en_base     | 50        | 50    | 9                | 80.0%       | 8.9              |
| sonar            | en_lottery  | 50        | 50    | 9                | 68.0%       | 10.2             |
| sonar            | en_terse    | 50        | 50    | 8                | 82.0%       | 8.0              |
| sonar            | en_integers | 50        | 50    | 12               | 42.0%       | 12.7             |

## B Deployment record

Table 4: Deployment record per engine. All calls ran on 1 September 2026, 09:05 to 09:08 UTC, one user message per call with a fresh context. No temperature, seed, or top-p parameters were sent; provider-side defaults for decoding are undisclosed by the endpoints and recorded as unknown. No search parameters were sent to any endpoint; sonar performs retrieval by default. The returned model id is the id string in each API response, uniform across all 200 calls per engine.

| engine           | requested model id   | returned model id    | endpoint host                     | extra parameters |
|------------------|----------------------|----------------------|-----------------------------------|------------------|
| gpt-5.6-luna     | gpt-5.6-luna         | gpt-5.6-luna         | api.openai.com                    | none             |
| claude-sonnet-5  | claude-sonnet-5      | claude-sonnet-5      | api.anthropic.com                 | max_tokens=128   |
| gemini-3.7-flash | gemini-3.7-flash     | gemini-3.7-flash     | generativelanguage.googleapis.com | none             |
| grok-4.5         | grok-4.5             | grok-4.5             | api.x.ai                          | none             |
| mistral-large    | mistral-large-latest | mistral-large-latest | api.mistral.ai                    | none             |
| sonar            | sonar                | sonar                | api.perplexity.ai                 | none             |

## References

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- [6] Žatuchin D (2026) Repeated queries exhaust an LLM’s brand recommendations but not its sources. Companion note, in the same repository